

Empirical Formula Problems

- Determine the empirical formula for each of the following compounds.
 - 85.7% carbon and 14.3% hydrogen
 - 52.9% aluminum and 47.1% oxygen
 - 62.6% lead, 8.4% nitrogen, and 29.0% oxygen
- A compound of carbon, hydrogen and chlorine consists of 49.0% carbon, 2.75% hydrogen and 48.3% chlorine by mass. What is the empirical formula of the compound?
- Chemical analysis of rubbing alcohol shows that it consists of 59.97% carbon, 13.35% hydrogen and 26.68% oxygen by mass. What is the empirical formula of rubbing alcohol?
- Analysis of a compound of potassium, sulfur and oxygen have the following results:
K 41.02% S 33.69% O 25.29%
What is the empirical formula of this compound?
- A 3.000 g sample of a compound was analyzed and was found to consist of 0.853 g Na, 0.962 g Cr, and 1.185 g of O. Determine the empirical formula of this compound.
- Caffeine is a stimulant found in coffee, tea, cola drinks and chocolate. Analysis of caffeine shows that it consists of 49.48% carbon, 5.19% hydrogen, 28.85% nitrogen and 16.48% oxygen by mass. Determine the empirical formula of caffeine.
- During the operation of a car battery, lead sulfate forms on the battery plates. Analysis of this compound shows that it consists of 68.3% lead, 10.6% sulfur, 21.1% oxygen by mass. What is the empirical formula of lead sulfate?

Answers

- 1a) CH_2 1b) Al_2O_3 1c) PbN_2O_8 2) $\text{C}_3\text{H}_2\text{Cl}$ 3) $\text{C}_3\text{H}_6\text{O}$
 4) $\text{K}_2\text{S}_2\text{O}_3$ 5) Na_2CrO_4 6) $\text{C}_4\text{H}_5\text{N}_2\text{O}$ 7) PbSO_4

Molecular Formula Problems

- Chemical analysis of one of the constituents of gasoline showed that it consists of 92.29% carbon and 7.71% hydrogen by mass if the molar mass of the compound is 78 g/mol, determine its molecular formula.
- A compound of silicon and fluorine was analyzed and found to consist of 33.0% silicon and 67.0% fluorine by mass. The molar mass of the compound was determined by experiment to be 170 g/mol. What is the molecular formula of the compound?
- Glucose is an important source of energy for organisms. Analysis of a sample of glucose shows that it consists of 39.95% carbon, 6.71% hydrogen and the remainder oxygen. In a separate experiment, the molar mass of glucose is found to be 180 g/mol. Determine the molecular formula of glucose.
- Lactic acid is the substance responsible for the taste of sour milk. Analysis of a sample of lactic acid shows that its percentage composition by mass is 40.00% carbon, 6.71% hydrogen and 53.29% oxygen. If the molar mass of lactic acid is found to be 90g, determine the molecular formula of lactic acid.
- A sample of a liquid used in dry cleaning was found to consist of 10.06% carbon, 89.10% chlorine, with the remainder being hydrogen. The molar mass of the compound was determined to be 119.6 g/mol. What is the molecular formula of the compound?

Answers

- 1) C_6H_6 2) Si_2F_6 3) $\text{C}_6\text{H}_{12}\text{O}_6$ 4) $\text{C}_3\text{H}_6\text{O}_3$ 5) CHCl_3